Graph-Aware Deep Reinforcement Learning for Adaptive Capacity Planning in Distributed Personalized Regenerative Medicine Manufacturing Networks

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Abstract

Autologous cell therapy and other personalized regenerative medicine (PRM) products are manufactured on demand across distributed clinical networks, where each product is *perishable*, *identity-bound* to a single patient and non-fungible, and needed only while that patient remains clinically eligible. These characteristics couple capacity, inventory, patient-indexed logistics, and patient outcomes into one networked decision. We formulate distributed PRM capacity planning as a constrained graph Markov decision process and release a reusable simulation testbed with patient-condition dynamics, expiry-aware inventory, geographic transfer delays, identity-preserving pre-manufacturing routing of intact specimen lots between qualified facilities, deterministic return of finished product to the collection site, and sharing of fungible reagents and idle bioreactor capacity. To solve it, we develop an *advantage-filtered residual graph convolutional network–deep deterministic policy gradient* controller (AFR-GCN–DDPG). The study-specific AFR

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label denotes the complete anchor-plus-residual controller, in which a graph convolutional state encoder learns bounded corrections to a transparent two-period look-ahead policy; it is not a new actor–critic backbone. The numerical results currently reported are from earlier configurations in which specimen routing was disabled; they compare controller architectures and are not estimates of the value of routing. In a five-seed demand-prior-drift experiment, with 300 training episodes and 100 paired holdout replications per seed, AFR-GCN–DDPG reduced the total weighted objective by 0.016360% relative to MDL-2, with a two-level 95% confidence interval fully below zero. A matched flat residual DDPG policy also improved on MDL-2, while the graph-versus-flat interval crossed zero, so this replenishment-only experiment supports residual learning but not a graph-specific gain. After extending the residual action to geographic reagent and idle-capacity transfers, a three-seed abrupt-regional-shift experiment reduced the objective by 1.1337% relative to MDL-2 and by 1.1166% relative to a parameter-matched flat policy while improving completion service. A separate conservative online-TD3 screen produced a final graph policy that improved on MDL-2 and matched flat TD3; however, its pooled final-versus-frozen-pretraining interval crossed zero. The evidence therefore supports graph-aware anchored control under geographically structured non-stationarity, while the incremental contribution of online actor–critic updates remains an open attribution question.

Keywords: Graph convolutional networks, Deep reinforcement learning, Actor–critic methods, Capacity planning, Distributed manufacturing, Personalized regenerative medicine, Perishable inventory, Transshipment

1. Introduction

Distributed make-to-order manufacturing networks are increasingly important in engineering systems where products are individualized, production capacity is scarce, and service outcomes depend on time-sensitive sequential decisions. Personalized regenerative medicine (PRM), and autologous cell therapy in particular, is a demanding instance of this class. Each therapy is manufactured from an individual patient's own biological material, so production can begin only when that patient's specimen, an available bioreactor, and sufficient consumables are simultaneously present, and the finished product must be returned to the same patient within a limited viability window. Because the product is patient-specific, it cannot be pre-produced, pooled, or substituted, and the opportunity to buffer against uncertainty with inventory is fundamentally limited [1, 2]. Three properties therefore distinguish PRM capacity planning from conventional make-to-order or inventory control. The product is *perishable*, subject to a hard shelf-life and vein-to-vein deadline; it is *identity-bound*, tied one-to-one to a single patient and non-fungible across patients; and its demand is driven by *patient health*, since a patient whose condition deteriorates while waiting may become clinically ineligible before treatment can be delivered. These are not marginal concerns. Autologous CAR-T manufacturing has a turnaround measured in weeks, an estimated 4–7% of patients never receive their product because of manufacturing failure, and in multiple myeloma roughly one quarter of patients die while waitlisted for a BCMA-directed CAR-T production slot [3]. Discrete-event simulation links longer waits directly to survival, with predicted one-year mortality in large B-cell lymphoma rising from about 36% to 76% as vein-to-vein time increases from one to nine months [4]. With the CAR-T market projected to reach tens of billions of US dollars by the early 2030s [5], delivering perishable, patient-specific product before patients deteriorate is a consequential operational challenge.

These properties make capacity planning a coupled, network-level decision problem rather than a collection of independent local ones. Locating production near patients through distributed or point-of-care manufacturing shortens fulfillment time, but it also creates spatial dependencies. A shortage at one clinic can sometimes be relieved by relocating idle bioreactor capacity or transshipping consumables from connected facilities, and each such action changes the future state of the whole network [6, 7]. Identity binding constrains ownership and use, not physical location: reagents and idle bioreactor capacity are fungible, whereas each specimen lot and finished product remain attached to one named patient. An intact waiting specimen may therefore be routed to a qualified manufacturing facility without being pooled, split, substituted, or reassigned, and the resulting product must return to that patient's collection or infusion site. Coordination must account for this patient-indexed logistics pathway as well as fungible-resource sharing. The state of such a system is naturally relational, since each facility is a node whose value depends on its neighbors' queues and inventories and on the links along which shortages, slack, specimens, and disruptions propagate. A flat feature vector encodes local conditions but discards this topology, which motivates a graph-structured state representation as a matter of substance rather than convenience.

Reinforcement learning (RL) is a natural fit because the problem is sequential. An action taken in one period alters future inventories, capacity, waiting specimens, and service outcomes. Continuous-control actor–critic methods, including deep deterministic policy gradient (DDPG) [8], twin delayed DDPG (TD3) [9], soft actor–critic (SAC) [10], and proximal policy optimization (PPO) [11], are standard tools for such real-valued control. Graph convolutional networks (GCNs) [12] provide a permutation-equivariant relational inductive bias for encoding networked state. This machinery is by now well established, and a substantial literature already couples graph learning with reinforcement learning for supply-chain and resource-allocation problems [13, 14, 15, 16, 17]. We therefore do not claim novelty in combining graph representation with reinforcement learning.

What that literature does not model is the PRM problem itself. Prior graph-and-RL studies for networked operations concentrate on demand forecasting, inventory-only ordering, or generic enterprise resource allocation, and they represent demand as fungible, abstract requests [13, 15, 18, 19, 20]; none model a product that expires, production that must return to the same patient, or demand generated by patient deterioration while waiting. Reinforcement learning has been applied within cell-therapy manufacturing, but at the level of a single bioprocess rather than a network of clinics [2], and recent evidence cautions that learned policies do not uniformly dominate well-tuned operational heuristics, particularly under perishability and non-stationarity [21]. The gap we address is thus a problem gap rather than a method gap, namely perishable, identity-bound, patient-condition-driven capacity planning across a clinic network, together with a decision framework and simulation testbed built specifically for it. To the best of our knowledge no public benchmark exists for this setting, so the environment is itself a contribution.

We organize the study around five research questions. **(RQ1)** With patient-indexed specimen routing held admissible across policies, does a graph representation of the clinic network improve coordinated specimen, capacity, reagent, and replenishment decisions over flat-state control, and under which supply-disruption and forecast-error conditions is any improvement largest? **(RQ2)** Which actor–critic backbone (DDPG, TD3, SAC, or PPO) is best suited to this constrained continuous-control problem, and is DDPG alone adequate? **(RQ3)** Does explicitly modeling patient condition and product expiry change the optimal policy and expose failures of condition-blind planners? **(RQ4)** Does a condition- and expiry-aware temporal encoder, which retains memory of deterioration trajectories, improve on a static graph encoder? **(RQ5)** Do learned policies outperform strong look-ahead heuristics, not only weak myopic rules, under supply disruption, demand forecast error, and patient-condition stress?

To study these questions we develop, first, a simulation of a multi-site autologous cell-therapy network in which patient health conditions, patient-indexed specimen routing, and product and material expiry are first-class drivers of demand and feasibility, and second, an advantage-filtered residual GCN–DDPG (AFR-GCN–DDPG) controller that coordinates specimen routing, capacity, reagent, and replenishment decisions across the network. AFR is our shorthand for the complete controller design: a transparent MDL-2 anchor, a graph-aware bounded residual actor, evidence-filtered teacher targets, and independently calibrated deployment safeguards. It does not replace or rename DDPG. The graph component encodes each facility's operational and patient-condition state together with information from connected facilities into network-aware embeddings. A deterministic policy-gradient actor then learns a bounded, state-dependent correction to the anchor policy, and a feasibility projection converts the corrected action into implementable controls. Advantage-filtered distillation retains a nonzero correction target only when common-random-number look-ahead predicts an improvement over the anchor; it is a teacher-data construction and regularization step, not a second algorithm or deployed policy. Checkpoint and residual-scale calibration provide a deployment safeguard. DDPG is the canonical proposed instantiation and provides continuity with prior PRM capacity-planning work; graph-aware TD3, SAC, and PPO variants serve as controlled stability and backbone comparisons. We adopt a statistically disciplined evaluation protocol (multiple random seeds, paired Monte Carlo outcomes, and two-level confidence intervals), because single-seed, mean-only reporting can make deep reinforcement learning results appear stronger and more stable than they are [22, 23]. Consistent with the clinical stakes of the problem, we report patient-service outcomes such as eligibility and waiting alongside operating cost, since a lower average cost can mask clinically unacceptable delay.

The main contributions of this study are as follows.

- We formulate distributed capacity planning for *perishable, identity-bound, patient-condition-driven* PRM manufacturing as a constrained networked sequential decision problem, in which patient deterioration and product and material expiry are explicit determinants of demand, feasibility, and cost.

- We develop a simulation environment realizing this formulation for a multi-clinic autologous cell-therapy network, with patient-condition dynamics, expiry-aware inventory, patient-indexed and identity-preserving pre-manufacturing specimen routing, finished-product return, and fungible-resource sharing. Routing is an operational capability of the model rather than a claimed methodological contribution.

- We propose AFR-GCN–DDPG, an advantage-filtered residual graph controller that learns bounded corrections to a transparent look-ahead anchor under demand and disruption uncertainty, with persistent teacher regularization and independently calibrated deployment safeguards.

- We establish a paired multi-seed evaluation protocol for comparisons with strong look-ahead and isolated-facility heuristics, flat-state reinforcement learning, and graph-aware backbone variants, and report a five-seed DDPG confirmation, a parameter-matched graph-attribution study, and a conservative online-TD3 screen with two-level bootstrap confidence intervals.

- We draw design implications for intelligent decision support in distributed, patient-specific healthcare manufacturing and related time-critical, make-to-order engineering systems.

The remainder of this paper is organized as follows. Section 2 formulates the distributed PRM capacity planning problem, extended with patient-condition states and product and material expiry. Section 3 reviews personalized regenerative medicine operations, reinforcement- and graph-learning for networked operational decisions, and the method foundations of continuous-control reinforcement learning and graph neural networks, closing with the design implications that motivate our approach. Section 4 presents the graph-aware deep reinforcement learning framework. Section 5 describes the computational experiments and results. Section 6 concludes the study and identifies future research directions.

2. Distributed Network Manufacturing and Capacity Planning Problem

In this section, we formulate the capacity planning problem for distributed manufacturing systems as a production simulation model based on a Markov decision process (MDP). Suppose the distributed manufacturing system consists of N manufacturing facilities that produce one type of make-to-order PRM product. Each manufacturing facility has its own production capacity, including bioreactors and reagent inventory, and can purchase reagents from source material suppliers. Facilities receive patient-indexed specimens and initiate production when the matched specimen, an available bioreactor, and sufficient reagents are co-located. Reagent transshipment and idle-bioreactor relocation are allowed among qualified facilities. In addition, one intact waiting specimen lot may be routed before manufacturing without changing the patient–specimen association; pooling, splitting, substitution, cross-patient use, and in-process transfer remain prohibited, and finished product is returned to the same patient's collection site (Section 2.3). Demand arrivals and supplier disruptions are modeled as stochastic processes, implying that local supply may be either insufficient or excessive relative to local demand

Table 1: Notation used in the distributed manufacturing and capacity planning model.

Sets and parameters	
Γ	Planning horizon indexed by t.
T	Manufacturing time for one PRM product, in periods, where $T < \Gamma$.
N	Set of manufacturing facilities indexed by n.
$\mathcal{E}_S$	Qualified facility pairs along which an intact waiting specimen lot may be routed before manufacturing.
m_t^n	Products that begin manufacturing at facility n at epoch t.
d_t^n	Demand arrivals at facility n in epoch t.
State variables	
s_t^n	Patient-indexed specimens waiting for manufacturing at their current facility n at epoch t.
r_t^n	Available reagent units at facility n at epoch t.
$\mathbf{b}_t^n$	Bioreactor work-in-process vector $(b_t^{n,1}, b_t^{n,2}, \ldots, b_t^{n,T})$, where $b_t^{n,\tau}$ denotes the number of bioreactors at facility n with τ periods remaining at epoch t.
P_t^n	Maximum reagent purchase quantity for facility n for epoch $t+1$.
Decision variables	
w_t^n	Executed integer net flow of intact waiting specimen lots into (positive) or out of (negative) facility n at epoch t, with identity and network conservation enforced (Section 2.3).
e_t^n	Reagent units added to or subtracted from facility n at epoch t and received or removed before epoch $t+1$.
q_t^n	Idle bioreactors added to or subtracted from facility n at epoch t and received or removed before epoch $t+1$.
p_t^n	Reagent purchase quantity for facility n for epoch $t+1$.
Patient-condition and perishability	
H_j	Latent health index of patient j, drawn $H_j \sim \text{Uniform}(0,1)$ at enrollment.
τ_j	Deterioration epoch of patient j, drawn $\tau_j \sim \theta\,\text{Weibull}(\kappa)$.
κ, θ	Shape and scale of the Weibull deterioration-epoch distribution.
α_j	Number of epochs since patient j enrolled, including waiting and manufacturing time.

at different decision epochs. The notation used throughout this section is summarized in Table 1.

Figure 1 illustrates the basic structure of the distributed manufacturing problem using a two-facility example. Each facility maintains a patient–specimen queue whose members carry a deteriorating survival index, together with reagent inventory, bioreactor capacity, and perishable finished product. At each decision epoch, condition-driven patient demand and disruption-prone reagent replenishment arrive under uncertainty. When one facility experiences excess demand, insufficient reagents, or limited production capacity, the network may transship reagents, relocate idle bioreactors, or route an intact waiting specimen to a qualified manufacturing facility. The latter is patient-indexed logistics rather than sharing: identity, ownership, and the final infusion destination never change. This structure motivates the MDP formulation, where the state captures facility-level demand, patient condition, material location and transit, inventory, and capacity, and the action coordinates both patient-indexed routing and fungible-resource decisions.

For the N facilities, T epochs of manufacturing time are required to produce one of the PRM products, and the planning horizon consists of Γ epochs where $\Gamma > T$.

At epoch $t \in \{1, 2, \cdots, \Gamma\}$, the production status of the n-th facility for $n = 1, \cdots, N$ is expressed as a state vector $\mathbf{x}_t^n$, where

$$\mathbf{x}_t^n = \left(s_t^n, r_t^n, \mathbf{b}_t^n, P_t^n\right), \tag{1}$$

Based on the state variables at epoch t, the facilities need to take action to adjust their production capacity for epoch $t + 1$. We define the action

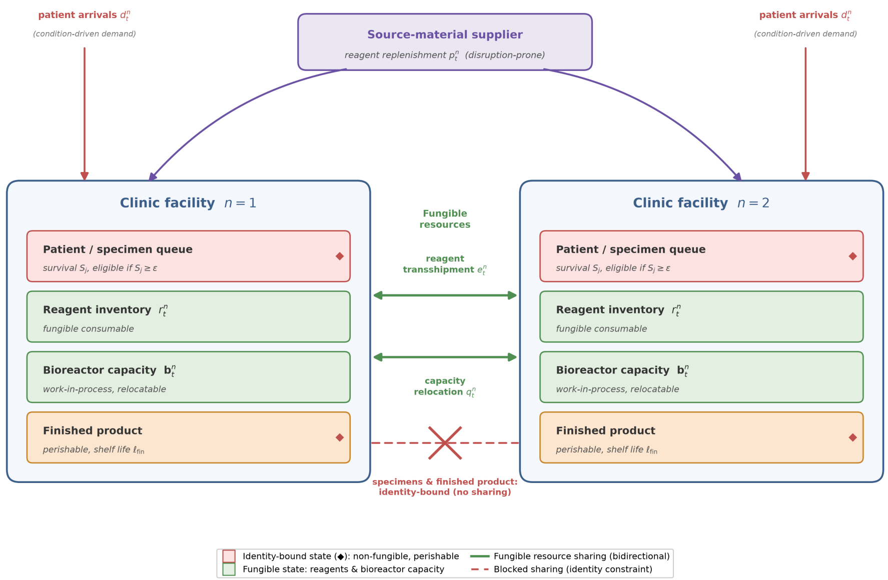

Figure 1: Illustration of a two-facility distributed manufacturing system for personalized regenerative medicine. Each facility holds a patient–specimen queue whose members carry a deteriorating survival index S_j (eligible while $S_j \geq \epsilon$), reagent inventory r_t^n, bioreactor capacity $\mathbf{b}_t^n$, and perishable finished product. Condition-driven patient demand d_t^n and disruption-prone reagent replenishment p_t^n arrive under uncertainty. Reagents and idle bioreactor capacity are fungible and can be rebalanced through transshipment e_t^n and relocation q_t^n (solid green links). The crossed red link denotes prohibited pooling, substitution, or cross-patient reassignment of specimens and products; it does not prohibit identity-preserving transport of one intact waiting specimen lot to a qualified manufacturing facility or return of its product to the same patient.

$\mathbf{a}_t^n$ as a vector that consists of the amount of capacity sharing and reagent replenishment. The action is decided at the beginning of the production period t and executed before the start of $t+1$.

$$\mathbf{a}_t^n = \left(w_t^n, e_t^n, q_t^n, p_t^n\right), \tag{2}$$

The goal of capacity planning is to minimize total production cost by dynamically selecting the actions from Equation (2) for all facilities, denoted by $\mathbf{A}_t = \left(\mathbf{a}_t^1, \cdots, \mathbf{a}_t^N\right)$, for each epoch $t \in \Gamma$. We define $V\left(\mathbf{X}_t\right)$ as the total future cost of the selecting actions $\mathbf{A}_t$ when the states from all facilities, denoted by $\mathbf{X}_t = \left(\mathbf{x}_t^1, \cdots, \mathbf{x}_t^N\right)$, are observed at epoch $t \in \Gamma$. The objective function can be expressed as a dynamic programming problem:

$$V\left(\mathbf{X}_t\right) = \min_{\mathbf{A}_t} \quad E\left[c\left(\mathbf{A}_t; \mathbf{X}_t\right) + \gamma V\left(\mathbf{X}_{t+1}\right)\right], \tag{3}$$

which is useful for finding the optimal action $\mathbf{A}_t$ for each state $\mathbf{X}_t$ at each epoch $t \in \Gamma$, where $c\left(\mathbf{A}_t; \mathbf{X}_t\right)$ is the immediate cost at time t, and $\gamma \in [0, 1]$ is a discount factor for balancing the impact of future costs in deciding actions. The choice of γ depends on the user's goal; if γ is closer to 1 (to 0), then it means the user cares more about long (short) term cost; if $\gamma = 0$, then only the cost for current epoch is considered. Equation (3) generalizes the optimality equation for the capacity planning problem in Li(2021) [6] in that cost functions $\mathbf{c}$ and $\mathbf{V}$ and the cumulative distribution function for the expectation operator can be unknown.

Equations (1)–(3) describe the base operational MDP shared with generic networked replenishment problems. The remainder of this section extends it along the three dimensions that distinguish PRM capacity planning, namely the deteriorating clinical condition of the patients awaiting treatment, the hard expiry of biological source material and finished product, and the autologous identity constraint that forbids pooling of patient-specific specimens. Section 4 then develops a graph-aware DRL approach that learns the cost function $\mathbf{c}$, the value function $\mathbf{V}$, and the expectation operator directly from simulated experience.

2.1. Patient condition and treatment eligibility

In generic capacity planning, a unit of unmet demand is deferred, backordered, or lost at a fixed penalty, and its value does not change while it waits. PRM demand behaves differently, because each arrival is a patient whose clinical condition deteriorates throughout the vein-to-vein pathway. A specimen that waits or travels too long, or a therapy whose matched patient deteriorates during manufacturing or product return, may no longer produce a deliverable treatment. We therefore attach an explicit condition process to every enrolled patient from queue entry through specimen transit, manufacturing, and product return, adapting the real-time patient-health modeling of Tseng et al. [24] to a distributed capacity-planning setting.

When a patient j enrolls at facility n, we draw a latent health index $H_j \sim \text{Uniform}(0, 1)$ and a deterioration epoch $\tau_j \sim \theta\,\text{Weibull}(\kappa)$ with shape κ and scale θ; H_j captures the heterogeneous baseline robustness across patients, while τ_j marks the onset of accelerated decline. Let α_j be the number of epochs since enrollment, including both queueing and manufacturing time. The patient's survival index $S_j \in [0, 1]$ (a normalized proxy for the likelihood that the patient remains treatable) evolves as

$$S_j(\alpha_j) = \ell(\alpha_j) + H_j\big(u(\alpha_j) - \ell(\alpha_j)\big), \tag{4}$$

where $u(\alpha) = \exp\big(-\lambda_H\, g(\alpha)\big)$ and $\ell(\alpha) = \exp\big(-\lambda_F\, g(\alpha)\big)$ are the survival trajectories of an ideally healthy and a maximally frail patient, with decay rates $\lambda_H < \lambda_F$, so that H_j interpolates each patient between the two bounds. The effective exposure time

$$g(\alpha) = \begin{cases} \alpha, & \alpha \leq \tau_j, \\ \tau_j + \rho\,(\alpha - \tau_j), & \alpha > \tau_j, \end{cases} \tag{5}$$

accelerates by a factor $\rho > 1$ once the patient passes the deterioration epoch τ_j, reflecting the clinically observed transition to rapid decline. A patient remains eligible for treatment only while $S_j \geq \epsilon$. Before production starts, crossing the threshold removes the patient and specimen from either the waiting or transit state. After production starts, the same condition process continues through manufacturing and any modeled product-return epochs. If the matched patient becomes ineligible before completion, the in-process therapy is discarded and the occupied bioreactor is cleaned during that epoch before returning to the idle pool at the next decision epoch. Treatment service is recorded only when the finished product reaches and is assigned to the original patient, not when production starts. Because condition erodes monotonically in expectation and accelerates after τ_j, waiting, specimen transit, manufacturing, and product-return delay are not neutral buffering decisions. Parameter values are reported in Section 5.

2.2. Perishability of materials and finished product

Both the biological source material and the finished product are perishable and identity-specific, so they cannot be stockpiled against future demand. A waiting or in-transit specimen that is not put into production within ℓ_{mat} epochs of enrollment expires, so the patient is lost and the reserved material is wasted. Finished product that is not delivered within ℓ_{fin} epochs of completion likewise expires and is discarded. We track, per facility n and epoch t, patients lost to waiting-stage ineligibility $\phi_t^{n,\mathrm{wait}}$, manufacturing-stage ineligibility $\phi_t^{n,\mathrm{mfg}}$, return-stage ineligibility $\phi_t^{n,\mathrm{ret}}$, local specimen expiry $\phi_t^{n,\mathrm{exp}}$, transit ineligibility $\phi_t^{n,\mathrm{tr\text{-}inel}}$, transit expiry $\phi_t^{n,\mathrm{tr\text{-}exp}}$, and modeled transit loss $\phi_t^{n,\mathrm{tr\text{-}loss}}$, together with finished units lost to expiry $\psi_t^{n,\mathrm{fin}}$. A manufacturing-stage ineligibility also creates one discarded in-process therapy. It is convenient to collect the total patients lost and total material wasted at each facility,

$$\begin{aligned}
\phi_t^n &= \phi_t^{n,\mathrm{wait}} + \phi_t^{n,\mathrm{mfg}} + \phi_t^{n,\mathrm{ret}} + \phi_t^{n,\mathrm{exp}} \\
&\quad + \phi_t^{n,\mathrm{tr\text{-}inel}} + \phi_t^{n,\mathrm{tr\text{-}exp}} + \phi_t^{n,\mathrm{tr\text{-}loss}}, \\
\psi_t^n &= \phi_t^{n,\mathrm{exp}} + \phi_t^{n,\mathrm{mfg}} + \phi_t^{n,\mathrm{tr\text{-}exp}} \\
&\quad + \phi_t^{n,\mathrm{tr\text{-}loss}} + \psi_t^{n,\mathrm{fin}},
\end{aligned} \tag{6}$$

where an expired or lost specimen contributes both a lost patient and a unit of wasted material. To signal impending losses before they occur, we also count the *at-risk unserved* patients whose survival has fallen within an urgency margin δ of the eligibility boundary, $\upsilon_t^n = \big|\{\, j \text{ waiting at } n : S_j < \epsilon + \delta \,\}\big|$.

2.3. Autologous identity constraint and admissible transfers

The autologous nature of PRM imposes a constraint with no analog in commodity supply chains. A specimen and the product manufactured from it are bound to one patient and cannot be substituted, pooled, split, or delivered to anyone else. Identity preservation does not, however, require the source material to remain at its collection site. We therefore allow *patient-indexed, identity-preserving, pre-manufacturing specimen routing*: only a complete lot in the waiting state may move, only along a qualified edge in $\mathcal{E}_S$, and at most once by a direct route before manufacturing begins. Multi-hop routing, cycles, in-production transfer, cross-patient consumption, and product reassignment are prohibited.

The aggregate decision w_t^n records the executed integer net flow produced by a patient-level routing executor. The executor pairs facilities with opposite signed requests, selects eligible waiting identities deterministically, and conserves lots so that $\sum_n w_t^n = 0$. Each selected patient's immutable identity record retains the collection facility, current material location, manufacturing facility, and transfer history. The main routing setting dispatches at epoch t, advances condition and specimen age once during one transit epoch, and makes the lot available before the epoch $t + 1$ production decision. After off-site manufacturing, the finished product remains assigned to the same patient and is returned to that patient's collection or infusion facility. Thus, routing changes manufacturing location without creating unrestricted specimen sharing.

2.4. Extended cost and the constrained graph MDP

The immediate cost augments the operational base cost c_{op} (covering reagent purchase, holding, and shortage; bioreactor holding and shortage; and reagent and capacity transfer) with three condition- and perishability-driven terms:

$$c(\mathbf{A}_t; \mathbf{X}_t) = c_{\mathrm{op}}(\mathbf{A}_t; \mathbf{X}_t) + \omega_{\mathrm{lost}} \sum_{n \in N} \phi_t^n + \omega_{\mathrm{exp}} \sum_{n \in N} \psi_t^n + \omega_{\mathrm{urg}} \sum_{n \in N} \upsilon_t^n, \qquad (7)$$

with weights ordered $\omega_{\mathrm{lost}} \gg \omega_{\mathrm{exp}} > \omega_{\mathrm{urg}}$ to encode the clinical priority, in which losing a patient dominates wasting material, which in turn dominates the shaping penalty for letting patients drift toward the eligibility boundary. The urgency term υ_t^n steers the controller to act pre-emptively (starting production or relocating capacity) before losses are actually incurred.

For clarity, we distinguish three quantities throughout the paper. For a policy π, the expected total weighted objective is

$$J_{\mathrm{tot}}^{\pi} = \mathbb{E}_{\pi}\left[\sum_{t=0}^{\Gamma-1} c(\mathbf{A}_t; \mathbf{X}_t)\right] = J_{\mathrm{op}}^{\pi} + J_{\mathrm{pat}}^{\pi}, \qquad (8)$$

where J_{op}^{π} accumulates reagent purchase, holding, and shortage; bioreactor holding and shortage; and reagent and capacity transfer, whereas J_{pat}^{π} accumulates the weighted patient-loss, expiry, and urgency terms in Equation (7). The total objective is the primary optimization and evaluation criterion. The two component groups are reported separately to show which decisions generate an improvement.

All three quantities are reported in *modeled weighted-objective units*. Resource-price coefficients can be interpreted as monetary equivalents under the adopted calibration, but the patient-loss, expiry, and urgency weights are penalty or shadow-value parameters rather than observed accounting expenditures. Consequently, an absolute difference in J_{tot} is not presented as realized cash savings without an external economic calibration. Some large components are driven partly by exogenous demand and patient deterioration and by structural constraints such as finite capacity and lead times. They are therefore only partly controllable, not “fixed” or wholly unavoidable.

For an anchor policy π_{anc} and candidate π, the headline relative improvement retains the complete objective as its denominator,

$$R_{\text{tot}} = \frac{J_{\text{tot}}^{\pi_{\text{anc}}} - J_{\text{tot}}^{\pi}}{J_{\text{tot}}^{\pi_{\text{anc}}}}. \tag{9}$$

We accompany it with absolute paired differences, component-specific relative changes, and clinical outcomes. When a valid oracle or lower-cost reference π_{ref} is available, we additionally report the fraction of anchor headroom captured,

$$R_{\text{headroom}} = \frac{J_{\text{tot}}^{\pi_{\text{anc}}} - J_{\text{tot}}^{\pi}}{J_{\text{tot}}^{\pi_{\text{anc}}} - J_{\text{tot}}^{\pi_{\text{ref}}}}. \tag{10}$$

This decomposition prevents a large patient-penalty or capacity-shortage component from obscuring a meaningful change in a directly affected operating component, while avoiding a post hoc replacement of the primary denominator.

To act on this cost, the controller needs visibility into the condition of each waiting queue. We therefore extend the facility state of Equation (1) with a condition summary $\boldsymbol{\sigma}_t^n$,

$$\tilde{\mathbf{x}}_t^n = \left(s_t^n, r_t^n, \mathbf{b}_t^n, P_t^n, \boldsymbol{\sigma}_t^n\right), \tag{11}$$

where $\boldsymbol{\sigma}_t^n$ contains the waiting count, waiting mean survival, near-expiry count, waiting survival-band histogram, specimens in transit to the facility, their mean survival, transferred specimens waiting at the facility, route-eligible waiting specimens, in-production count, in-production mean survival, and in-production at-risk count. The controller can therefore anticipate queue, transit, and manufacturing risk rather than merely react to aggregate demand.

Finally, we cast the extended problem as a *constrained graph MDP*. The facilities and their admissible information, reagent, capacity, and specimen-logistics links form a graph $\mathcal{G} = (N, \mathcal{E})$, with the qualified specimen edge set $\mathcal{E}_S \subseteq \mathcal{E}$ enforced separately from fungible-resource feasibility. The global state $\tilde{\mathbf{X}}_t = (\tilde{\mathbf{x}}_t^1, \ldots, \tilde{\mathbf{x}}_t^N)$ is a signal over the nodes of $\mathcal{G}$, and the action $\mathbf{A}_t$ assigns per-node purchases, fungible-resource transfers, and conserved patient-indexed specimen-routing requests. The optimality equation (3) carries over verbatim with the extended state $\tilde{\mathbf{X}}_t$ and cost (7), but the value function must now trade current operating and logistics cost against the future risk of waiting, transit, patient loss, and expiry. This constrained graph MDP (perishable, identity-bound, and patient-condition-driven) is the problem the remainder of the paper addresses, and its graph structure is precisely what the method of Section 4 is designed to exploit.

3. Literature Review

We review three bodies of work that our approach draws together, namely the operations of personalized regenerative medicine (PRM) and decentralized manufacturing; reinforcement learning and graph learning for networked operational decisions; and the underlying method foundations of continuous-control reinforcement learning and graph neural networks. Rather than surveying each exhaustively, we read them as an argument in which each stream supplies part of what a distributed PRM capacity-planning controller needs, none supplies all of it, and together they imply a small set of design choices that we make explicit at the end of the section. Throughout, we treat the learning machinery as established and concede that graph representation and reinforcement learning have already been combined for supply-chain problems; what remains open is the PRM problem itself, one that is perishable, identity-bound, and patient-condition-driven.

3.1. Personalized regenerative medicine and decentralized manufacturing

PRM supply chains differ substantially from conventional make-to-stock manufacturing systems because products are individualized, time-sensitive, and tightly coupled with patient-specific clinical pathways. In autologous cell therapy, patient material must be collected, transported, manufactured, quality-tested, and returned for administration, and manufacturing can begin only when a specimen, consumables, and production capacity are jointly available. These characteristics make fulfillment time, service reliability, and resource utilization central operational measures. Wang et al. [1] used simulation to compare centralized and point-of-care autologous cell therapy supply chain strategies and evaluated cost, fulfillment time, service level, and resource utilization. Li and White [6] further studied decentralized autologous cell therapy manufacturing networks and showed that transshipment and relocatable capacity can reduce the cost of resilience under supplier disruption. Most directly relevant to our patient-condition modeling, Tseng et al. [24]

integrated real-time patient health data into an autologous cell-therapy simulation and showed that representing patient condition, rather than treating demand as static, changes fulfillment and eligibility outcomes. We adopt this patient-condition perspective as a first-class driver of demand and feasibility, but embed it within a networked capacity-planning formulation solved by graph-aware learned control, rather than evaluating fixed policies in a simulation study alone.

These studies establish that PRM network design and inter-facility coordination are first-order engineering decisions rather than implementation details. Related work on cell therapy manufacturing process control also highlights the potential of reinforcement learning in uncertain, nonlinear, and partially observed manufacturing environments [2]. Together, this literature motivates a data-driven decision-support framework that can represent both the temporal dynamics of patient-specific manufacturing and the spatial dependencies created by distributed facilities.

3.2. Reinforcement learning for supply chain and manufacturing decisions

Reinforcement learning has been widely studied for sequential supply chain and manufacturing decisions. Early work applied RL to inventory replenishment in multi-echelon supply chains and perishable inventory systems [25, 26]. More recent studies have used deep reinforcement learning to handle high-dimensional state spaces, stochastic demand, and simulation-based decision environments. For example, Oroojlooyjadid et al. [27] developed a deep Q-network approach for the beer game, demonstrating that DRL can learn replenishment policies in decentralized, partially observed supply chain settings.

Recent supply chain RL studies have moved toward more realistic constraints and engineering settings. Bermudez et al. [19] proposed distributional constrained reinforcement learning for multi-period supply chain optimization with production and inventory constraints. Eisenach et al. [20] studied shared-resource capacity control in inventory management using a neural coordination mechanism and a reinforcement learning buying policy. Stranieri et al. [21] benchmarked classical and deep reinforcement learning inventory policies for pharmaceutical supply chains with perishability and non-stationary demand. These studies support the use of simulation-trained RL policies for uncertain operational control, but most focus on replenishment or inventory management rather than the joint capacity, reagent, and replenishment decisions (under a perishable, identity-bound demand process) that arise in distributed PRM manufacturing.

3.3. Structure-informed, residual, and baseline-relative reinforcement learning

Inventory-control evidence does not support treating generic end-to-end DRL as automatically superior to established operating rules. Gijsbrechts et al. [28] found that carefully tuned DRL can match state-of-the-art policies across lost-sales, dual-sourcing, and multi-echelon problems, while the invited roadmap of Boute et al. [29] argues that learning methods should exploit the structural policy knowledge accumulated in inventory research. De Moor et al. [30] made this principle operational by using strong perishable-inventory heuristics as teacher policies for reward shaping, improving training speed, stability, and, in suitable instances, performance relative to the teacher. Hybrid decomposition offers another route: Stranieri et al. [31] assigned production decisions to DRL and shipping decisions to stochastic programming, obtaining more reliable performance than pure DRL. Non-stationary-demand experiments further show the value of conditioning policies on forecast information rather than assuming a fixed arrival law [32].

Residual reinforcement learning provides the most direct methodological foundation for our controller. The residual formulation decomposes the executed action into a competent base-controller action and a learned correction, reducing the part of the control law that must be discovered from experience [33]. Staessens et al. [34] showed in an industrial-control setting that bounding the correction relative to a conventional controller can preserve its stabilizing role while adapting to uncertain operating conditions. Baseline-relative policy-improvement methods likewise motivate accepting a learned update only when there is credible evidence of improvement over the incumbent policy [35]. In stochastic inventory control, Deep Controlled Learning combines approximate policy iteration with common random numbers and statistically efficient candidate selection to overcome the difficulty generic DRL has in beating strong heuristics [36]. These ideas motivate our MDL-2 anchor, bounded graph residual, common-random-number advantage filtering, and held-out deployment calibration. Our validation fallback is an empirical deployment safeguard, however, and should not be interpreted as inheriting the formal monotonic-improvement or closed-loop stability guarantees proved under the assumptions of those methods.

3.4. Deep reinforcement learning for regenerative medicine capacity planning

The closest antecedent to this work is Tseng et al. [7], who developed a deep reinforcement learning approach for dynamic capacity planning in decentralized regenerative medicine supply chains. Their study demonstrated that simulation-trained policies can adapt to demand and supply uncertainty without requiring exact prior knowledge of the underlying stochastic processes. This is directly aligned with the objective of using data-driven policy learning for PRM capacity planning.

The present study builds on this line of work by adding an explicit graph representation layer to encode inter-facility relationships. In decentralized PRM networks, facilities are linked through reagent transshipment, idle-bioreactor relocation, and qualified patient-indexed specimen logistics, while every specimen and finished product remains bound to one patient. Therefore, a facility's decision depends not only on its own local state but also on the states, route-eligible queues, transit pipeline, and fungible-resource conditions of connected facilities. A flat vectorized state representation can include all facility variables, but it does not explicitly exploit the graph structure that governs how information, material, and resources propagate across the network. This motivates integrating graph convolutional representation learning with continuous-control actor–critic reinforcement learning.

3.5. Multi-agent reinforcement learning and graph learning for networked supply chains

Recent multi-agent reinforcement learning studies are relevant because distributed manufacturing networks naturally involve decentralized information and coordinated actions. Mousa et al. [18] formulated decentralized inventory control as a multi-agent problem and studied centralized-training/decentralized-execution policies under local information constraints. Leluc et al. [37] similarly demonstrated the promise of cooperative multi-agent reinforcement learning for inventory management under stochastic demand and lead times. These studies show that decentralization is not only an application feature but also an algorithmic design dimension.

In parallel, graph neural networks have emerged as a natural representation tool for supply chains and other networked operational systems. Chang et al. [14] learned production functions on supply-chain graphs using temporal graph neural networks and an inventory module, showing the value of explicit relational structure for supply-chain inference and forecasting. Ahn et al. [15] used attention-based graph neural networks for probabilistic supply and inventory prediction in enterprise supply-chain networks, while Ahn et al. [16] combined graph neural networks, offline reinforcement learning, and policy simulation for dynamic supply planning. Kotecha and del Rio Chanona [13] provide the closest graph-control antecedent: their GNN–MARL framework dynamically parameterizes heuristic inventory policies and uses graph structure to coordinate decentralized supply-chain agents. AFR-GCN–DDPG differs by learning a bounded continuous correction around a fixed look-ahead anchor, filtering correction targets through common-random-number look-ahead, and controlling patient-condition, capacity, fungible-resource, and patient-indexed specimen-routing decisions in an identity-bound PRM network rather than inventory replenishment alone. These studies collectively suggest that graph-based learning is useful for networked supply chain systems; however, most existing work remains focused on prediction, inventory-only control, or general supply-chain planning rather than continuous-action control of PRM-specific capacity, reagent, and specimen flows.

3.6. Method foundations: continuous-control reinforcement learning and graph neural networks

The actor proposes real-valued pressures for capacity relocation, reagent replenishment and transfer, and specimen routing; a deterministic executor converts those pressures into feasible mixed continuous and integer decisions. This points to continuous-control actor–critic methods coupled with explicit projection. Deterministic policy gradients and DDPG [8] extended actor–critic learning to continuous action spaces, but are known to suffer from value overestimation and training instability. TD3 [9] mitigates overestimation through clipped double-Q learning, delayed policy updates, and target smoothing; SAC [10] adds entropy regularization for more robust exploration and stability; and PPO [11] is a widely used on-policy alternative built on a clipped surrogate objective. The practical implication is that DDPG, though historically the default for continuous supply-chain control, is no longer an uncontested choice. A defensible study should compare backbones rather than commit to one, and should expect the more stable off-policy methods to reduce the run-to-run variance that plagues DDPG.

Graph neural networks encode relational structure through neighborhood aggregation. The graph convolutional network [12] and its successors (GraphSAGE [38] for inductive neighborhood sampling and graph attention networks [39] for learned edge weighting) provide a permutation-equivariant inductive bias that a flat multilayer perceptron lacks. Coupling such encoders with reinforcement learning has become a recognized paradigm for networked decision problems [17]. For a clinic network whose facilities interact through patient-indexed specimen logistics, capacity relocation, and consumable transshipment, this relational bias is a modeling choice of substance rather than convenience, because it allows a single policy to reason over the topology along which patients, shortages, slack, and disruptions propagate.

3.7. Evaluation considerations for deep reinforcement learning

Evaluation rigor is especially important for DRL-based engineering applications. Henderson et al. [22] showed that deep reinforcement learning results can be sensitive to random seeds, hyperparameters, and implementation details. Agarwal et al. [23] further argued that few-run comparisons can be statistically unreliable and recommended more robust aggregate metrics and interval estimates. These findings imply that a DRL-based supply chain study should report implementation details, training budgets, out-of-sample test procedures, and variability across replications or random seeds when possible. In the present study, we therefore emphasize common simulation settings across all benchmark policies, detailed implementation parameters for the graph-aware actor–critic agents, and Monte Carlo evaluation under multiple disruption and demand forecast scenarios, with interquartile-mean performance and confidence intervals reported across random seeds.

3.8. Research gap and design implications

Read together, these literatures imply five design choices for a distributed PRM capacity-planning controller. First, because the decisions are sequential and real-valued, the controller should use continuous-control actor–critic reinforcement learning rather than value-based or discretized methods. Second, because a facility's value depends on its network position and resources

Table 2: Positioning of this study relative to closely related literature.

Literature stream	Representative studies	Main limitation relative to this work
PRM operations	Simulation and decentralized planning [1, 6, 7]	No learned graph policy combining patient condition, perishability, identity, and coordinated continuous control.
Supply-chain RL	Inventory and constrained control [27, 19, 21, 20]	Primarily inventory or replenishment, not coupled PRM capacity and fungible-resource sharing.
Decentralized MARL	Cooperative inventory control [18, 37]	Not tailored to identity-bound, patient-specific make-to-order manufacturing.
Supply-chain GNNs	Graph forecasting and graph-aware planning [14, 15, 16, 13]	Mostly predictive or inventory-focused rather than continuous PRM network control.
Structured residual RL	Heuristic-guided and baseline-relative learning [30, 34, 35, 36]	No graph-structured PRM application with patient-condition risk and identity constraints.

propagate along inter-facility links, its state should be encoded with a graph rather than a flat vector. Third, because strong operational rules already solve much of the nominal problem, the learned policy should target bounded, evidence-supported corrections to a transparent anchor rather than relearn the complete policy from scratch. Fourth, DDPG provides a natural canonical formulation and continuity with earlier PRM capacity-planning work, but its known instability requires controlled comparisons with TD3 and secondary SAC/PPO variants rather than an untested assumption that DDPG is always best. Fifth, because reinforcement learning results are sensitive to seeds and implementation details, evaluation should report multi-seed interquartile means with confidence intervals and, given the clinical stakes, patient-service outcomes such as eligibility and waiting.

None of these streams, however, addresses the problem that defines our setting. PRM operations research motivates adaptive, networked capacity planning but does not learn relational policies. Supply-chain graph-and-reinforcement-learning research supplies the relational-control machinery but treats demand as fungible and abstract, targeting forecasting or inventory-only control. Cell-therapy reinforcement learning operates on a single bioprocess rather than a clinic network. As a result, no prior work models the three defining properties of PRM capacity planning (a perishable product with a hard viability deadline, an identity-bound product that cannot be pooled or substituted across patients, and demand driven by patient deterioration while waiting) within a single graph-aware continuous-control formulation. Our contribution closes this gap. We cast the problem as a constrained graph MDP, build a reusable simulation testbed that realizes it, and study a family of graph-aware actor–critic controllers, pairing a GCN state encoder with DDPG, TD3, SAC, and PPO, against strong flat-state and heuristic baselines.

4. Graph-Aware Deep Reinforcement Learning Framework

Overview: a shared graph encoder with interchangeable actor–critic backbones

Distributed manufacturing networks for personalized regenerative medicine (PRM) comprise complex, interconnected facilities whose operating states are best represented as graphs rather than flat vectors. Classical reinforcement learning that treats each facility as independent cannot capture these spatial dependencies. We therefore adopt a two-part design shared by every method we study. A graph convolutional network (GCN) encoder maps the networked state to permutation-equivariant, neighborhood-aware node embeddings, composed with a continuous-control actor–critic backbone that maps those embeddings to feasibility-projected capacity, reagent, and replenishment actions. Writing the controller as *encoder* $\circ$ *backbone* lets us hold the relational representation fixed while swapping the reinforcement-learning backbone, so that any performance difference between backbones is attributable to the learning algorithm rather than to the state representation.

Primary AFR-GCN–DDPG controller and backbone variants

The primary proposed controller is advantage-filtered residual GCN–DDPG, abbreviated AFR-GCN–DDPG. The acronym names the complete controller rather than a new actor–critic backbone: *residual* means that the DDPG actor modifies, rather than replaces, the MDL-2 anchor; *advantage-filtered* means that nonzero teacher corrections are retained only when paired common-random-number look-ahead estimates a positive improvement over that anchor. This filtering is a teacher-data construction and regularization step inside AFR-GCN–DDPG, not a separate deployed policy. The controller combines the relational inductive bias of a GCN with the deterministic continuous-control mechanism used in earlier PRM capacity-planning research, while reducing the burden of relearning a complete operational policy from scratch. Following residual and constrained-residual control [33, 34], the actor learns only a bounded correction to a transparent look-ahead anchor. We also evaluate TD3, SAC, and PPO behind the same GCN encoder (Table 4) as controlled backbone variants: TD3 through clipped double-Q learning, delayed policy updates, and target smoothing; SAC through entropy-regularized stochastic policies; and PPO through an on-policy clipped surrogate objective. All four share the graph representation, feasibility-projection layer, environment, and evaluation protocol, so backbone comparisons do not erase AFR-GCN–DDPG's role as the canonical proposed instantiation. A GCN encodes relational structure but has no temporal memory of its own; the actor–critic component supplies the sequential decision dynamics, and the two together encode spatial correlations and temporal trade-offs jointly.

We therefore present the proposed AFR-GCN–DDPG controller in full (the anchor-plus-residual action, GCN encoder, actor–critic updates, advantage-filtered supervision, and feasibility projection) and then note where each stability-oriented variant modifies the critic target, policy parameterization, or update schedule; the graph encoder and feasibility-projection layer are identical across backbones.

Table 3: Terminology for the proposed controller and its matched comparisons.

Term	Meaning and role in this study
GCN; DDPG; TD3; SAC; PPO	**Established acronyms.** Standard graph-encoding and actor–critic algorithms, used with their conventional meanings and cited source formulations.
MDL-2	**Study baseline.** Two-period mean-demand look-ahead heuristic and transparent anchor; the label is not a general reinforcement-learning acronym.
Advantage-filtered distillation	**Training step.** Common-random-number look-ahead retains a nonzero teacher correction only when it improves on the anchor. It is not a separate policy.
AFR	**Study-specific prefix.** Shorthand for the complete *advantage-filtered residual* design: anchor, bounded learned correction, teacher regularization, feasibility projection, and deployment calibration.
AFR-GCN–DDPG	**Proposed controller.** AFR with a GCN state encoder and DDPG actor–critic backbone.
AFR-Flat-DDPG; AFR-GCN–TD3	**Matched comparisons.** The first removes the graph encoder; the second changes the RL backbone. Neither is a separate proposed framework.

Table 4: Continuous-control actor–critic backbones evaluated behind the shared GCN encoder.

Backbone	Policy class	Sampling	Key mechanism and role in this study
DDPG [8]	Deterministic	Off-policy	Canonical backbone of the proposed AFR-GCN–DDPG controller; retains continuity with prior PRM capacity-planning work.
TD3 [9]	Deterministic	Off-policy	Stability-enhanced variant using clipped double-Q learning, delayed policy updates, and target smoothing.
SAC [10]	Stochastic	Off-policy	Entropy-regularized objective for more robust exploration and training stability.
PPO [11]	Stochastic	On-policy	Clipped surrogate objective; an on-policy alternative to the off-policy backbones.

Figure 2 shows the graph abstraction used by the proposed graph-aware framework. Each healthcare clinic is represented as a node with facility-specific state features, including waiting and inbound specimens and their condition summary, reagent inventory, available bioreactor capacity, and other local operational conditions. The edges encode different forms of inter-facility dependency. Information-flow edges let the model propagate operational and patient-condition signals among neighboring clinics; resource-sharing edges represent feasible reagent transshipment relationships; capacity-sharing edges connect each clinic to the central capacity hub; and an explicitly configured subset of qualified inter-clinic links permits identity-preserving routing of intact waiting specimen lots. By applying graph convolution over this structure, the model learns a network-aware representation in which each facility's decision is informed not only by its own state but also by the states of connected facilities. This graph embedding is then passed to the actor–critic backbone to support coordinated specimen-routing, capacity, reagent, and replenishment decisions.

Formally, let $G_t = (V, E_t)$ represent the manufacturing network at epoch t, where each node $v_i \in V$ denotes a facility with state features x_i^t and edge weights e_{ij}^t indicate logistic or transshipment costs. The GCN encodes local and neighborhood information using a layer-wise propagation rule:

$$h_i^{(l+1)} = \sigma\Big(W^{(l)} \sum_{j \in \mathcal{N}(i)} \frac{1}{c_{ij}} h_j^{(l)}\Big), \tag{12}$$

where $h_i^{(0)} = x_i^t$, $W^{(l)}$ are trainable weights, $\sigma(\cdot)$ is a ReLU activation, and c_{ij} is a degree-based normalization factor. The resulting embedding $H_t = \{h_i^{(L)}\}_{i=1}^N$ captures the current global state of the network.

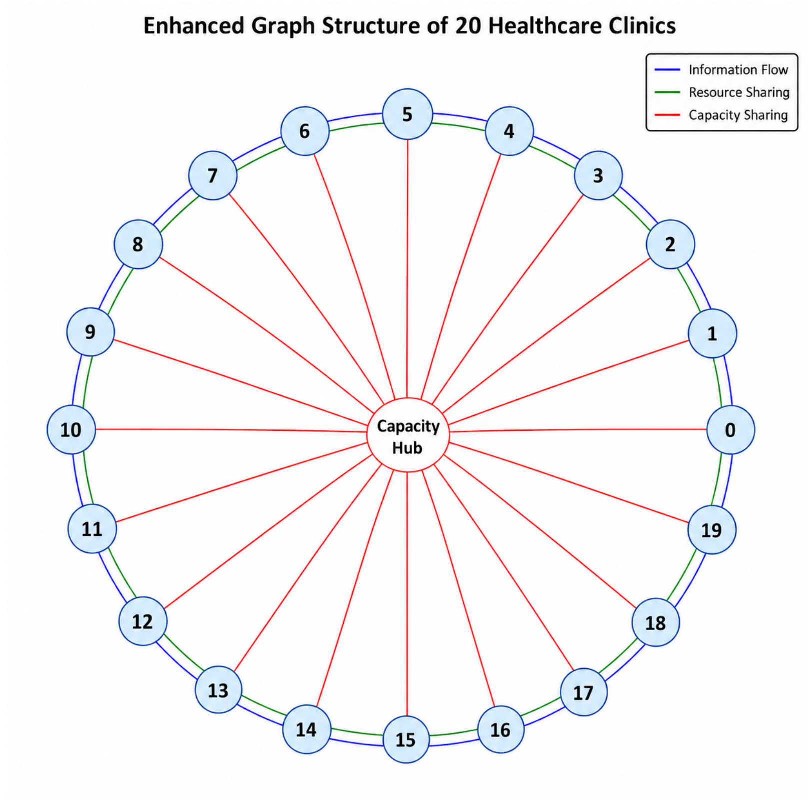

Figure 2: Graph representation of a distributed personalized regenerative medicine manufacturing network with 20 healthcare clinics. Each outer node represents a clinic or manufacturing facility, and the central node represents the shared capacity pool. Blue edges indicate information flow among neighboring clinics, green edges indicate fungible-resource relationships, and red radial edges indicate capacity sharing between each clinic and the central capacity hub. Qualified specimen-logistics links are configured as a constrained subset of inter-clinic relationships and are omitted from this conceptual edge legend. The graph convolutional network encodes local and neighboring state for downstream patient-indexed routing and resource decisions.

Overview of the graph-aware actor–critic workflow (DDPG instantiation)

The method initiates with a dual initialization of the actor and critic networks of the backbone and the GCN for graph state processing. Given an observed state X_t, represented as raw graph data reflecting the current status of the manufacturing network, the GCN is used as a preprocessing unit. It transforms this raw graph into a condensed representation X_t', which retains the critical structural information of the original graph.

Following this transformation, the DDPG algorithm gets involved. Using the graph-processed state X_t', the actor-network selects an action, which is based on the actor–critic architecture. This action could range from decisions regarding resource allocation to inventory replenishments. Once executed within the manufacturing environment, the system provides a feedback loop in the form of a reward, essentially evaluating the decision's efficacy.

The DDPG then uses this feedback, in tandem with the replay buffer, to refine both the actor and critic networks, ensuring better decision-making in subsequent iterations. The GCN, on its part, continuously updates its understanding of the graph structure as the manufacturing network evolves, ensuring it remains adaptive and accurate in its representations.

Initialization

- Critic network $Q(X_t, A_t|\theta_Q)$ initialized with weights θ_Q.
- Actor network $\mu(X_t|\theta_\mu)$ initialized with weights θ_μ.
- GCN initialized with weights ω for processing graph states.

DDPG Component

The DDPG agent operates on the GCN-encoded state H_t to generate a bounded residual action $\Delta A_t = \mu(H_t|\theta_\mu)$. Let A_t^{anc} denote the action produced by the transparent look-ahead anchor and let ρ be a vector of decision-group-specific trust-region scales. The raw proposed action is

$$A_t^{\text{raw}} = A_t^{\text{anc}} + \rho \odot \Delta A_t. \tag{13}$$

This residual parameterization preserves the anchor's basic inventory logic while allowing the GCN policy to adapt patient-indexed specimen routing, replenishment, and fungible resource-sharing decisions to network-wide patient pressure, geographic connectivity, disruptions, and demand-prior drift. The critic network $Q(H_t, A_t|\theta_Q)$ estimates the expected cumulative cost. The optimization follows the standard DDPG updates:

$$y_i = c_i(A_i; H_i) + \gamma Q'(H_{i+1}, \mu'(H_{i+1}|\theta_{\mu'})|\theta_{Q'}), \tag{14}$$

$$L_Q = \frac{1}{N}\sum_i (y_i - Q(H_i, A_i|\theta_Q))^2, \tag{15}$$

$$\nabla_{\theta_\mu} J = \frac{1}{N}\sum_i \nabla_a Q(H_i, a|\theta_Q)|_{a=\mu(H_i)} \nabla_{\theta_\mu}\mu(H_i|\theta_\mu). \tag{16}$$

Advantage-filtered residual initialization and deployment

The useful residual corrections in AFR-GCN–DDPG are sparse because the MDL-2 anchor is already competitive in many states. We therefore initialize the residual actor through advantage-filtered teacher construction. At a sampled state, we compare a small set of decision-group residual candidates, including specimen routing and network-resource actions, with the zero-residual MDL-2 action over a finite look-ahead horizon. Candidate and anchor rollouts use common random numbers so that their score difference is not dominated by different demand, deterioration, or disruption realizations. A nonzero target is retained only when it improves the configured operating-cost and patient-outcome score; otherwise the target is the zero residual. To keep the sparse improvement labels from being overwhelmed, improving and zero-residual examples receive equal total weight. The weighted advantage-filtered data remain in the actor loss during DDPG fine-tuning rather than being used only as a one-time warm start.

Deployment uses two independent validation stages. First, checkpoints saved during training are compared on common validation scenarios. Second, the selected checkpoint is evaluated over a fixed set of residual trust-region scales, including zero. The nonzero residual is deployed only when it improves validation cost without reducing service; otherwise the controller falls back to MDL-2. This mechanism keeps the deployed controller within the residual GCN–DDPG family while preventing a poorly generalized correction from overriding the operational anchor.

Reward function

The agents optimize the negative of the immediate cost defined in Section 2.4. For each epoch t, the reward is

$$r_t = -\, c(\mathbf{A}_t; \tilde{\mathbf{X}}_t), \tag{17}$$

where $c(\cdot)$ is the extended cost of Equation (7), comprising the operational base cost c_{op} (reagent purchase, holding, and shortage; bioreactor holding and shortage; specimen and reagent logistics; and capacity relocation) together with the patient-loss, material-and-product-expiry, and urgency terms. Optimizing this reward therefore requires the agent to trade current operating and logistics cost against the future risk of patient losses and expiry, rather than resource utilization alone.

Action decoding and feasibility projection.. Because the actor network outputs continuous residuals, an additional deterministic decoding and feasibility-projection step is used before actions are executed in the simulation environment. The raw anchor-plus-residual action A_t^{raw} is first scaled to the admissible range of each decision variable and then clipped to its variable-specific lower and upper bounds. The specimen block expresses signed facility pressure rather than anonymous divisible flow. It is scaled by the maximum route quantity and rounded half away from zero to integer requests; a shared executor then matches opposite requests only across qualified edges and selects specific eligible patient identities. Reagent transshipment, bioreactor relocation, and replenishment are decoded through their corresponding bounds and feasibility rules.

The decoded action is subsequently passed through a repair operator that enforces the operational constraints of the manufacturing network. First, nonnegativity constraints are imposed so that the resulting action cannot remove more reagents, idle bioreactors, or route-eligible waiting specimen lots from a facility than are available at that epoch. Second, replenishment quantities are capped by the maximum purchasable reagent quantity P_t^n and by supplier availability. Third, network conservation is enforced separately for fungible resources and integer specimen lots. A specimen route is legal only for a waiting identity that has not previously moved; each executed

Algorithm 1 Graph-aware actor–critic for distributed PRM capacity planning (DDPG instantiation; TD3/SAC/PPO swap the critic target, policy parameterization, or update schedule).

Require: Anchor policy π_{anc}, residual candidates $\mathcal{D}$, look-ahead horizon L, discount factor γ, soft-update rate τ, actor and critic learning rates α_μ, α_Q, exploration scale σ, minibatch size N.

1: Initialize GCN encoder g_ω, actor $\mu(\cdot \mid \theta_\mu)$, critic $Q(\cdot, \cdot \mid \theta_Q)$, target networks μ' and Q', and replay buffer $\mathcal{R}$.

2: Build weighted teacher data by comparing $\pi_{\text{anc}}(X) + \delta$, $\delta \in \mathcal{D}$, with $\pi_{\text{anc}}(X)$ under common-random-number L-step rollouts; use target δ^* only when it improves the score and zero otherwise.

3: Fit the residual actor to the weighted advantage-filtered targets and retain this loss as an actor regularizer.

4: **for** each environment step t **do**

5: Observe raw graph state X_t and compute graph embedding $Z_t = g_\omega(X_t)$.

6: Select exploratory residual action $A_t = \pi_{\text{anc}}(X_t) + \rho \odot \mu(Z_t \mid \theta_\mu) + \epsilon_t$, where $\epsilon_t \sim \mathcal{N}(0, \sigma^2 I)$.

7: Project or clip A_t to the feasible set and execute the resulting action $\hat{A}_t$.

8: Observe cost/reward r_t, next state X_{t+1}, and store $(X_t, \hat{A}_t, r_t, X_{t+1})$ in $\mathcal{R}$.

9: **if** an update is scheduled **then**

10: Sample minibatch $(X_i, A_i, r_i, X_{i+1})_{i=1}^N$ from $\mathcal{R}$ and compute $Z_i = g_\omega(X_i)$, $Z_{i+1} = g_\omega(X_{i+1})$.

11: Compute targets $y_i = r_i + \gamma Q'(Z_{i+1}, \mu'(Z_{i+1} \mid \theta_{\mu'}) \mid \theta_{Q'})$.

12: Update critic by minimizing $L_Q = N^{-1} \sum_i (y_i - Q(Z_i, A_i \mid \theta_Q))^2$.

13: Update actor using the deterministic policy gradient plus the weighted advantage-filtered regularizer.

14: Backpropagate through the GCN encoder and softly update target networks.

15: **end if**

16: **end for**

17: Select checkpoint and residual scale on independent validation scenarios; return the calibrated residual policy or the MDL-2 fallback.

donor event has one matched receiver event, and unmatched requests are recorded as blocked rather than repaired by substitution. Finally, all transfer and routing actions are restricted to their configured graph edges. Finished-product return is a deterministic identity-preserving lifecycle transition, not an actor decision.

Equivalently, the executed action can be interpreted as the feasible action $\hat{A}_t$ obtained by projecting the decoded actor output onto the feasible action set $\mathcal{F}(X_t)$:

$$\hat{A}_t = \Pi_{\mathcal{F}(X_t)}(A_t^{\mathrm{raw}}) \, , \tag{18}$$

where $\mathcal{F}(X_t)$ is determined by current facility states, patient lifecycle and route history, inventory availability, capacity availability, supplier limits, the autologous identity constraint, and graph connectivity. This projection step improves engineering credibility by ensuring that all actions executed in the simulation correspond to implementable specimen-routing, capacity, replenishment, and transshipment decisions.

Each stability-oriented backbone reuses this encoder and projection layer unchanged and alters only the reinforcement-learning update. TD3 replaces the single critic with clipped double-Q targets, delays actor updates, and adds target-policy smoothing; SAC replaces the deterministic actor with an entropy-regularized stochastic policy and a soft value target; and PPO replaces off-policy replay with on-policy rollouts optimized under a clipped surrogate objective. In all cases the actor and critic operate on the GCN embedding Z_t rather than on the flat state, and the executed action is the feasibility-projected $\hat{A}_t$.

Summary

The proposed method is therefore a residual GCN–DDPG controller: a GCN encoder supplies a permutation-equivariant, network-aware state representation; a deterministic actor learns bounded corrections to a transparent look-ahead anchor; and a feasibility-projection layer guarantees that every executed action respects inventory, capacity, supplier, patient identity, route history, and connectivity constraints. The routing-primary architecture comparison holds specimen-routing availability, the anchor, residual action space, advantage-filtered teacher protocol, projection, training budget, and evaluation seeds fixed between GCN and parameter-matched flat encoders. The earlier backbone comparison instead holds the graph encoder fixed while replacing DDPG with TD3. Residual SAC and PPO are secondary family extensions because their stochastic residual distributions and likelihood calculations require additional matched implementation work. These variants test backbone robustness without redefining them as separate proposed methods.

5. Computational Experiments

This section reports three empirical stages designed to separate anchor improvement, graph representation, and online reinforcement-learning attribution. The first stage calibrates patient-condition dynamics and evaluates replenishment residuals under demand-prior drift. The second expands the action space to geographically structured reagent and idle-capacity transfers and compares the graph policy with a parameter-matched flat policy. The third holds the pretrained graph controller fixed while testing whether conservative online TD3 updates add value beyond frozen pretraining. The experiments compare the learned graph-aware correction policy with its heuristic anchor, other heuristic and look-ahead policies, flat-state residual policies, and graph-aware backbone variants under the same simulation environment.

5.1. Objective

The objective of this study is to compare the proposed residual GCN–DDPG framework with traditional heuristic methods, its MDL-2 anchor, flat-state residual DDPG, and stability-oriented graph-aware actor–critic variants in distributed manufacturing.

5.2. Methodology

5.2.1. Benchmarking Setup

The final benchmark is organized as a hierarchy of matched comparisons rather than a single undifferentiated algorithm ranking (Table 5). The first comparison tests whether reinforcement learning improves the MDL-2 anchor. The second isolates the graph representation by replacing the GCN with a flat MLP while keeping the entire residual-learning recipe fixed. The third tests whether the result is specific to DDPG by repeating the residual comparison with TD3. Online rollout shields and the supervised GCN shield selector are reported separately because the former uses additional simulation at decision time and the latter is not an RL policy. Pure end-to-end RL and the stochastic SAC/PPO family are reported as secondary comparators rather than being allowed to substitute for these mandatory tests.

The heuristic baselines are:

- **MYO (Myopic):** This heuristic minimizes single-period cost without considering future demand.
- **MDL-1 (One-period Mean Demand Lookahead):** MDL-1 extends the planning horizon to one lookahead period using deterministic state transitions based on mean demand.
- **MDL-2 (Two-period Mean Demand Lookahead):** MDL-2 extends the planning horizon to two lookahead periods using deterministic state transitions based on mean demand.
- **ISO (Isolated Regional Facilities):** ISO assumes no resource sharing between facilities and focuses only on local replenishment decisions.
- **F-MYO (Forecast Myopic):** F-MYO augments current-period network balancing with the demand forecast available at the decision epoch.
- **uMYO and pMYO (Patient-aware myopic policies):** uMYO adds urgency-based replenishment and capacity surges, while pMYO uses a bounded patient-risk workload uplift. These controls prevent the proposed method from being compared only with condition-blind heuristics.

The learning-based comparisons are:

- **MLP residual DDPG / flat-state residual DDPG:** This baseline uses the same MDL-2 anchor, residual action space, DDPG actor–critic structure, training budget, and feasibility projection as the proposed method, but removes the GCN encoder. Instead, all facility-level states are concatenated into a flat vector and passed directly into multilayer perceptron actor and critic networks. This baseline is designed to answer whether performance improvements come from reinforcement learning alone or from explicitly encoding graph-structured inter-facility dependencies.

- **Residual GCN–TD3 and matched flat residual TD3:** These policies keep the anchor, advantage-filtered supervision, action constraints, training data, and evaluation streams fixed while replacing the DDPG update with TD3. They test whether twin critics, delayed actor updates, and target smoothing improve stability.

- **Pure GCN and flat actor–critic policies:** From-scratch DDPG and TD3 quantify the benefit of anchoring; SAC and PPO provide secondary off-policy stochastic and on-policy comparisons. Residual SAC/PPO are included only after their residual-space probability calculations and matched flat variants pass the same validation protocol.

In addition, graph ablation experiments are planned to evaluate the role of different edge types. Specifically, we consider removing capacity-sharing edges and removing resource-sharing edges to assess whether these relational structures contribute to policy performance under disruption and demand uncertainty.

All methods receive the same decision-time observations and are evaluated on identical paired random-number streams. In demand-prior-drift experiments, heuristics receive the same misspecified demand prior available

Table 5: Final comparison hierarchy for the proposed controller.

Comparison layer	Required policies	Question answered
Anchor improvement	AFR-GCN–DDPG versus MDL-2	Does the learned correction improve the strong policy from which it starts?
Strong heuristics	ISO, MYO, F-MYO, uMYO, pMYO, MDL-1, and MDL-2	Is the controller competitive with sharing, forecast-aware, patient-aware, and look-ahead rules?
Representation ablation	AFR-GCN–DDPG versus matched AFR-flat-DDPG	Is any gain attributable to graph representation rather than residual learning alone?
Backbone robustness	Residual GCN–DDPG versus residual GCN–TD3, each with a matched flat counterpart	Does the result persist when the deterministic actor–critic backbone changes?
Anchor ablation	Residual versus pure GCN/flat DDPG and TD3	Does the heuristic anchor improve sample efficiency, stability, and final performance?
Deployment reference	MDL-2/pMYO rollout shields and supervised GCN shield selector	How does policy quality trade off against online simulation cost, and can RL outperform non-RL distillation?
Secondary family	GCN/flat SAC and PPO; residual forms after matched implementation	Is the conclusion robust to stochastic off-policy and on-policy learning?
Graph mechanism	Full graph, no/shuffled edges, and geographic-edge ablations	Does the policy use topology rather than merely additional parameters?

to the controller and no method receives the realized future demand rates. Learned policies use the same training-scenario distribution, environment-interaction budget, checkpoint-selection budget, and held-out evaluation seeds within each matched comparison. Online decision time and the number of simulator rollouts are reported separately for rollout-based methods. This prevents a performance difference from being created by unequal forecasts, simulator access, evaluation noise, or hidden online computation.

5.2.2. Performance Metrics

We evaluate each method using both objective-based and engineering-relevant performance metrics. The primary metric is the average total weighted objective J_{tot} in Equation (8). It includes the complete operating-cost group (reagent purchase, holding, and shortage; bioreactor holding and shortage; and reagent and capacity transfer) and the patient-related penalty group (patient loss, material or therapy expiry, and urgency). The percentage improvement reported in headline comparisons always uses the anchor's complete J_{tot} as the denominator, as in Equation (9); we do not remove large components after observing the result.

Because a weighted total can be dominated by patient-loss or capacity-shortage penalties, every formal comparison also reports the absolute paired difference and an additive cost decomposition. Component-specific percentages are secondary mechanism measures: they reveal whether a policy saves purchase, holding, shortage, transfer, or patient-penalty cost, but do not replace the primary total-objective comparison. When an oracle or other defensible lower-cost reference is evaluated, we also report the headroom-capture measure in Equation (10). To strengthen the application relevance of the experiments, we track the following secondary metrics:

- **Cost decomposition:** operating/base cost and its reagent, bioreactor, and transfer components, together with patient-loss, expiry, and urgency penalties.
- **Completion service level:** the fraction of eligible patient demand completed within the planning horizon.
- **Patient loss and manufacturing-period ineligibility:** the number of patients who become ineligible or expire before treatment and the fraction who become ineligible after manufacturing begins.
- **Average waiting time / fulfillment delay:** the average number of epochs that patient specimens wait before manufacturing begins.
- **Reagent shortage frequency:** the frequency with which manufacturing is delayed because insufficient reagents are available.
- **Bioreactor utilization:** the fraction of available bioreactor capacity used for production across facilities.
- **Routing, transshipment, and relocation activity:** patient-indexed specimen route count, distance, transit time, route cost, and blocked requests, together with reagent movements and idle-bioreactor relocations. These are mechanism diagnostics rather than evidence that routing is itself a contribution.
- **Inference time:** the computational time required for a trained policy or heuristic to generate one epoch-level decision.

These metrics are included because the objective of PRM capacity planning is not only to reduce a modeled weighted objective, but also to support timely patient-specific manufacturing, effective resource utilization, and reliable system performance under demand and supply uncertainty. Cost-weight sensitivity is therefore required before translating an objective difference into an economic claim.

5.2.3. Heuristic Implementations

We provide brief descriptions of the heuristic methods implemented:

- **MYO (Myopic):** MYO focuses on minimizing costs within the current period without considering future consequences.
- **MDL-1 and MDL-2:** These methods extend the planning horizon to one and two lookahead periods, respectively, using deterministic state transitions based on mean demand.
- **ISO (Isolated Regional Facilities):** ISO assumes no sharing of resources between manufacturing facilities and primarily concentrates on reagent replenishment.

5.2.4. Implementation and statistical protocol

The models are implemented in Python using PyTorch, and all experiment parameters, training seeds, checkpoint candidates, deployment scales, and evaluation streams are stored in configuration files. Within a matched comparison, the graph and flat policies receive the same state variables, anchor actions, teacher targets, environment interactions, validation budget, and common-random-number (CRN) holdout episodes. The graph policy replaces the flat encoder with message passing over the clinic graph; the remaining actor–critic and deployment components are held fixed or parameter matched.

The replenishment-residual DDPG and TD3 confirmation uses 300 training episodes for each of five seeds, checkpoint candidates between episodes 50 and 300, and 100 independent paired holdout replications per seed. Reported confidence intervals use a two-level bootstrap over training seeds and paired evaluation replications. The regional network-action experiment uses three independently initialized graph and flat policies, 30 paired validation episodes, and 100 previously unseen paired holdout replications per seed. Because its selected checkpoint is dominated by advantage-filtered teacher distillation, that experiment is interpreted as graph-policy evidence rather than as independent evidence for online DDPG updates.

The conservative online-TD3 screen starts from the corresponding frozen pretrained graph and flat checkpoints and runs 100 online episodes for seeds 0, 1, and 2. Each run completed 5,200 update calls, including 5,073 critic updates and 2,286 delayed actor updates. Final and frozen checkpoints are evaluated with identical CRN keys, using 20 validation and 100 holdout replications per seed for the final policies and 5 validation and the same 100 holdout replications per seed for the frozen controls. This campaign was run on an NVIDIA RTX 4090 GPU; all optimization diagnostics were finite, and no CPU fallback occurred. These experiment-specific budgets replace the earlier generic one-million-step description and make the reported results traceable to the executed configurations.

The routing-primary campaign is prospectively separated from those earlier no-routing results. It holds patient-indexed specimen routing enabled for every primary policy and compares GCN residual DDPG, a parameter-matched flat residual DDPG controller, and the MDL-2 routing anchor. One fresh routing-enabled teacher is shared across the learned architectures. After two five-episode seed-0 smoke runs pass identity, CUDA, update, finite-loss, checkpoint, parameter-gap, and nonzero-route gates, the locked pilot runs six independent 100-episode jobs in the order GCN seeds 0–2 followed by flat seeds 0–2. Final and frozen-pretraining checkpoints are evaluated on four routing scenarios with 100 paired CRN replications per training seed; lead-zero specimen transit and one-epoch product return are prespecified sensitivities. The primary attribution is GCN versus matched flat, GCN versus MDL-2, and final versus frozen pretraining. No-routing is retained only for deterministic mechanics regression or a separately approved supplementary ablation, not as a default formal training arm. No numerical result from this routing campaign is reported until its locked outputs pass the preregistered gates.

5.3. distributed manufacturing simulation

To evaluate the proposed method, we now construct a simulation environment of a decentralized manufacturing system consisting of $N = 20$ manufacturing facilities and its dedicated reagent supplier. Each PRM product requires $T = 3$ epochs of manufacturing time, and the considered finite problem horizon contains $\Gamma = 52$ decision epochs. For every epoch t, there will be d_t^n new demand for facility n. The demand arrival process for facility n follows a Poisson distribution Poisson (λ_n). The supplier disruption follows independent Bernoulli processes Ber (ζ_n), i.e., the facility n has a probability

Algorithm 2 State normalization

Require: State variables: s_t^n, r_t^n, b_t^n, and p_t^n

Maximal values of the state variables for all facilities: M_s, M_r, M_b, and M_p

Ensure: Normalized state variables: $\hat{s}_t^n$, $\hat{r}_t^n$, $\hat{b}_t^n$, $\hat{p}_t^n$

for $n = 1$ to N **do**

  $\hat{s}_t^n = s_t^n / M_s$

  $\hat{r}_t^n = r_t^n / M_r$

  $\hat{b}_t^n = b_t^n / M_b$

  $\hat{p}_t^n = p_t^n / M_p$

end for

ζ_n to have supplier disruption and could not purchase any reagent for the epoch. This simulation can be further adjusted to reflect any real situation, e.g., supply disruption and demand uncertainty, that may be encountered. Such flexibility in using the simulation system will be demonstrated in Section 5.4 when we compare GCN-DDPG with other methods.

Patient-condition parameters are calibrated before policy comparison. The nominal scenario uses $\lambda_H = 0.0013$, $\lambda_F = 0.0156$, Weibull shape $\kappa = 1.5$ and scale $\theta = 4$, post-deterioration acceleration $\rho = 3$, eligibility threshold $\epsilon = 0.80$, and waiting-time pressure $0.0013\alpha^2$. We selected these values from a fixed-seed sweep whose pre-specified target was a 10–15% *patient ineligibility during manufacturing* rate over the three-week process. This outcome is not labeled a technical manufacturing failure: it records a patient who becomes clinically ineligible after production begins, whereas a process or release-specification failure would be a separate stochastic event. The target is consistent with the order of magnitude of pre-infusion attrition reported for autologous CAR-T pathways [3] and with a 5% weekly eligibility-transition benchmark, which gives $1 - (1 - 0.05)^3 = 14.26\%$ over three weeks.

The patient-facing weights are $\omega_{\mathrm{lost}} = 500{,}000$, $\omega_{\mathrm{exp}} = 100{,}000$, and $\omega_{\mathrm{urg}} = 25{,}000$. The earlier exploratory values $(50{,}000, 40{,}000, 5{,}000)$ made one-time patient loss cheaper than even one period of combined reagent and bioreactor shortage, so a policy could lower its objective when a patient left the system. The calibrated ordering removes this directional inconsistency and is held fixed for all policy comparisons.

The epoch is assumed to occur weekly. Reagent and idle-bioreactor transfers enter an in-transit pipeline and arrive after one, two, or three epochs, according to whether the geographic distance is at most 500 miles, between 500 and 1,500 miles, or above 1,500 miles. In the routing-primary campaign, an intact waiting specimen may take one direct qualified route. The main setting uses one transit epoch and immediate modeled return of finished product to the collection facility; zero-epoch specimen transit and one-epoch product return are sensitivities. The previously reported case studies set $w_t^n = 0$ and are therefore not routing-effect experiments. After demand and any due transfers or routes have arrived, the next observed state is $\mathbf{x}_{t+1}^n = \left(s_{t+1}^n, r_{t+1}^n, \mathbf{b}^n, P_{t+1}\right)$ for facility $n = 1, \ldots, N$, according to Equation (1). The state variables are updated by the following rules:

$$
\begin{aligned}
s_{t+1}^{n} &= s_t^n - m_t^n + d_t^n + w_t^n \\
r_{t+1}^{n} &= r_t^n - m_t^n + e_t^n + p_t^n \\
b_{t+1}^{n,\tau} &= \begin{cases} b_t^{n,0} - m_t^n + b_t^{n,1} + q_t^n & \text{if } \tau = 0 \\ b_t^{n,\tau+1} & \text{if } 1 \le \tau \le T-2 \\ m_t^n & \text{if } \tau = T-1 \end{cases}
\end{aligned}
$$

where $m_t^n = \min\left(s_t^n, b_t^{n,0}, r_t^n\right)$ is the number of products that begin manufacturing at epoch t, d_t^n is the demand arriving at facility n in epoch t, and P_{t+1} follows an exogenous Markov chain from P_t. The operating component of the single-period objective can be decomposed as

$$
c_{\text{op}}\left(\mathbf{A}_t; \mathbf{X}_t\right) = \sum_{n=1}^{N} C_{\text{op},t}^{n},
$$
$$
C_{\text{op},t}^{n} = C_{\text{Reg},t}^{n} + C_{\text{Bio},t}^{n} + C_{\text{Tran},t}^{n}.
$$

The meanings of $C_{\text{Reg},t}^n$, $C_{\text{Bio},t}^n$, and $C_{\text{Tran},t}^n$ are summarized below. These terms constitute only c_{op}; the patient-related terms in Equation (7) must also be included to obtain the total weighted objective.

- $C_{\text{Reg},t}^n = c_R p_t^n + h_R\left(r_{t+1}^n - s_{t+1}^n\right)^+ + p_R\left(s_{t+1}^n - r_{t+1}^n\right)^+$ is the cost related to reagent at epoch t, including the reagent replenishment cost $c_R p_t^n$, the reagent overstock holding cost $h_R\left(r_{t+1}^n - s_{t+1}^n\right)^+$, and the reagent understock penalty $p_R\left(s_{t+1}^n - r_{t+1}^n\right)^+$;

- $C_{\text{Bio},t}^n = h_B\left(b_{t+1}^{n,0} - s_{t+1}^n\right)^+ + p_B\left(s_{t+1}^n - b_{t+1}^{n,0}\right)^+$ is the cost related to bioreactor at epoch t, including the bioreactor overstock holding cost $h_B\left(b_{t+1}^{n,0} - s_{t+1}^{n}\right)^{+}$, and the bioreactor understock penalty $p_B\left(s_{t+1}^{n} - b_{t+1}^{n,0}\right)^{+}$;

- $C_{\mathrm{Tran},t}^{n} = K_R\left|e_t^n\right| + K_B\left|q_t^n\right| + K_S\left|w_t^n\right|$ is the logistics-cost expression. The routing-primary campaign includes the patient-indexed specimen term; it vanishes only in the earlier no-routing case studies.

Table 6: Modeled cost coefficients used in the calibrated experiments. Values are weighted-objective units and are held fixed across policies. Transfer coefficients are base values before the geographic distance and travel-time surcharge. Patient-related coefficients are penalty or shadow-value parameters rather than observed cash expenditures.

Component	Coefficient	Interpretation
Reagent purchase, c_R	42,174.0	per purchased unit
Reagent holding, h_R	113.5	per excess unit-period
Reagent shortage, p_R	86,504.5	per shortage unit-period
Bioreactor holding, h_B	14.4	per idle unit-period
Bioreactor shortage, p_B	50,273.6	per shortage unit-period
Reagent transfer, K_R	200.0	per transferred unit, before surcharge
Capacity transfer, K_B	500.0	per relocated unit, before surcharge
Patient loss, ω_{lost}	500,000	per lost patient
Expiry, ω_{exp}	100,000	per wasted material or therapy unit
Urgency, ω_{urg}	25,000	per at-risk unserved patient-period

5.4. Case Study 1: Interconnected Clinic Network Analysis

This case study holds the demand prior equal to the arrival distribution and varies the supplier disruption rate over 0.05, 0.30, and 0.60. Its purpose is to measure the value of graph-aware coordination when the deterministic heuristics are not disadvantaged by demand misspecification. The final matched multi-seed campaign for these three regimes is still pending; consequently, we do not report numerical disruption-effect claims in the present draft. This case study will use the same five training seeds, paired Monte Carlo evaluation, checkpoint selection, and two-level bootstrap protocol as Case Study 2.

5.5. Case Study 2: Adaptability to Non-Stationary Demand

This case study evaluates the main robustness hypothesis under patient-condition and geographic network dynamics. The 20 clinics are partitioned into four five-clinic demand groups. MDL-2 plans with Poisson prior rates $[7.5, 3.2, 6.0, 3.5]$, whereas the simulator generates arrivals using true rates $[10.5, 3.52, 7.5, 3.15]$. The environment also includes patient deterioration, expiry, clustered demand shocks, regional supplier disruptions, and distance-dependent transfer lead times. During training, demand rates are randomized between 0.8 and 1.5 of the scenario rates and forecast error between 0 and 0.35; evaluation is fixed to the scenario above.

Before retraining learned policies, we evaluated ISO and MDL-2 with 100 paired Monte Carlo replications to verify that the calibrated patient process and objective produce the intended clinical range. Table 7 reports this calibration pilot and the subsequent matched learned-policy screen. Learned-policy results produced before the complete patient lifecycle and objective calibration are not comparable and are therefore retired rather than carried forward into this experiment.

Both heuristics fall inside the pre-specified clinical calibration band: manufacturing-

Table 7: Calibrated demand-prior-drift results. ISO and MDL-2 use the initial 100-replication calibration stream; learned-policy and matched heuristic rows aggregate 100 paired replications for each of three training seeds. Objective values are the total modeled weighted objective in billions of cost units, not realized accounting expenditure. Lower objective, patient loss, and manufacturing-period ineligibility are better; higher completion service and eligibility are better.

Method	Objective (10^9 units)	Completion	Eligibility	Lost	Mfg. inelig.
Initial ISO calibration	3.990456	0.337612	0.377537	4021.92	0.146398
Initial MDL-2 calibration	3.895694	0.371948	0.421024	3698.87	0.121364
AFR-Flat-DDPG	3.927584	0.370359	0.507030	3730.36	0.122223
AFR-GCN-DDPG	3.927594	0.370359	0.507030	3730.35	0.122223
AFR-GCN-TD3	3.927628	0.370358	0.507030	3730.36	0.122223
MDL-2	3.927620	0.370359	0.507030	3730.36	0.122223
ISO	4.011547	0.337521	0.460044	4043.10	0.146468
pMYO	4.180485	0.343311	0.496776	3894.09	0.133332
MYO	4.194549	0.342889	0.496586	3897.01	0.133582

period patient ineligibility is 14.64% for ISO and 12.14% for MDL-2. MDL-2 reduces the mean total weighted objective by 94.76 million modeled cost units (2.37%) relative to ISO; the paired 95% interval for MDL-2 minus ISO is $[-108.81, -80.71]$ million cost units, and MDL-2 has a lower objective in 92 of 100 paired replications. It also loses 323.05 fewer patients on average, discards 49.40 fewer in-process therapies, and raises completion service by 0.03434.

The calibrated learned-policy screen used 100 training episodes and three training seeds. AFR-GCN–DDPG selected and deployed a nonzero residual in all three seeds; AFR-GCN–TD3 and AFR-Flat-DDPG each returned to MDL-2 in one seed. AFR-GCN–DDPG reduced the mean total weighted objective relative to MDL-2 by only 26,519 modeled cost units (0.000675%), with a two-level bootstrap 95% interval of $[-89{,}588, 40{,}330]$ cost units. Its mean objective was 9,460 cost units higher than the matched flat residual policy, with interval $[-7{,}951, 24{,}621]$. Completion service and patient loss were effectively unchanged. Thus the screen demonstrates safeguard-compatible parity with a strong anchor, but it does not support either a material RL improvement or a graph-representation advantage. The accepted correction primarily substitutes additional reagent purchase and holding cost for lower reagent shortage cost. Because this residual changes replenishment only, the next matched experiment expands the bounded correction to distance-aware reagent and idle-bioreactor transfers, where geographic message passing can affect the decision.

The small headline percentage in this screen is intentionally computed against the complete weighted objective rather than a narrower, post hoc denominator. Patient-related penalties and capacity-shortage penalties can dominate that total, while a replenishment-only residual directly changes only a subset of the operating components. We therefore interpret 0.000675% as statistical parity rather than as a material economic saving. Subsequent network-residual comparisons report the total-objective change together with operating, bioreactor-shortage, transfer, and patient-penalty decompositions so that denominator dilution and the mechanism of any improvement remain visible.

5.5.1. Formal five-seed demand-prior-drift confirmation

The formal replenishment-residual comparison increased the training budget to 300 episodes and used five independent training seeds. Each selected checkpoint was evaluated on 100 paired holdout replications, yielding 500 paired outcomes per method. Table 8 reports the mean outcomes, and Table 9 reports paired differences. The learned-policy values in these tables supersede the 100-episode diagnostic screen in Table 7.

Table 8: Five-seed demand-prior-drift confirmation. Objective values are the total modeled weighted objective in billions of units. Each row aggregates 100 paired holdout replications for each of five training seeds.

Method	Objective (10^9 units)	Completion	Eligibility	Waiting	Lost	At-risk unserved
MDL-2	1.268257	0.429982	0.530313	3.052625	3665.59	5206.41
AFR-Flat-DDPG	1.268089	0.430320	0.530523	3.052012	3663.28	5203.86
AFR-GCN–DDPG	**1.268049**	**0.430321**	**0.530524**	**3.052012**	**3663.27**	**5203.86**
AFR-GCN–TD3	1.268241	0.430317	0.530523	3.052019	3663.30	5203.88

Table 9: Paired five-seed demand-prior-drift comparisons. Differences are candidate minus baseline, so a negative objective difference favors the first method. Confidence intervals use the two-level bootstrap over training seeds and paired replications.

Comparison	ΔJ_{tot} (units)	Relative gap	Paired 95% CI	Wins / 500
AFR-GCN–DDPG vs. MDL-2	−207,485	−0.016360%	[−332,641, −91,367]	323
AFR-Flat-DDPG vs. MDL-2	−167,610	−0.013216%	[−286,899, −46,230]	319
AFR-GCN–DDPG vs. AFR-Flat-DDPG	−39,876	−0.003145%	[−89,702, 14,025]	285
AFR-GCN–TD3 vs. MDL-2	−15,823	−0.001248%	[−142,607, 99,437]	268
AFR-GCN–DDPG vs. AFR-GCN–TD3	−191,662	−0.015112%	[−234,659, −148,401]	377

AFR-GCN–DDPG improves on MDL-2 in every training-seed mean and has a two-level confidence interval fully below zero. Completion service increases by 0.000340, eligibility by 0.000211, average waiting time falls by 0.000612, patients lost fall by 2.32, and at-risk unserved patients fall by 2.55. This supports a statistically stable anchor improvement under demand-prior drift, but not a large economic effect. The matched flat policy also improves on MDL-2, and the direct graph-versus-flat interval includes zero. Replenishment residual learning is therefore supported in this scenario, whereas an independent graph-representation advantage is not.

The matched TD3 policy preserves the patient-facing improvements but has a cost interval crossing zero. DDPG is lower cost than TD3 in all five training-seed means, and their paired interval is fully below zero. This comparison supports DDPG as the canonical backbone for the proposed replenishment-residual controller; it is not a claim that DDPG generally dominates TD3 outside this residual-learning design.

5.6. Case Study 3: Graph Attribution Under Regional Non-Stationarity

This earlier no-routing case study combines patient-condition dynamics, clinic geography, distance-based transfer delays, regional demand heterogeneity, and demand regime changes. Unlike the replenishment-only experiment, the network residual contains a node head for signed replenishment corrections and edge heads for reagent and idle-bioreactor transfer corrections. Edge flows are projected to preserve resource conservation and feasibility, while patient-indexed specimen routing is disabled by that historical protocol. This action space gives graph message passing a direct route to affect geographically structured decisions.

5.6.1. Network AFR-GCN–DDPG under an abrupt regional shift

The abrupt-shift policy uses transfer-focused advantage-filtered targets and a fixed deployment scale selected on 30 paired validation episodes. Three independently initialized graph policies and parameter-matched flat policies were then evaluated on the same 100 previously unseen paired holdout replications per seed. The graph and flat models contain 524,775 and 528,404 trainable parameters, respectively, so the comparison does not give the graph policy a parameter-count advantage.

Table 10: Three-seed abrupt-regional-shift comparisons. Objective differences are in millions of modeled weighted-objective units. Each comparison contains 300 paired holdout outcomes.

Comparison	ΔJ_{tot} (10^6 units)	Relative gap	Paired 95% CI	Δ completion	Δ patients lost
Network AFR-GCN–DDPG vs. MDL-2	-25.39	-1.1337%	$[-29.76, -20.95]$	$+0.001357$	-0.70
Network AFR-GCN–DDPG vs. AFR-Flat-DDPG	-25.01	-1.1166%	$[-29.35, -20.61]$	$+0.002836$	-9.86

All three seed-level objective means favor the graph controller. Relative to MDL-2, the completion-service 95% interval is $[0.000240, 0.002458]$ and the patients-lost interval is $[-6.42, 5.07]$; aggregate patient-loss noninferiority therefore passes, but the strict per-seed clinical gate passes for two of three seeds. Relative to the matched flat policy, the completion interval is $[0.001766, 0.003898]$ and the patients-lost interval is $[-15.24, -4.31]$. This is the current statistically significant evidence for a graph-specific benefit: it appears when the action space includes geographically structured edge decisions, not when the actor modifies replenishment alone.

Table 11 explains why the total-objective percentage is smaller than the change in the component directly affected by the policy. MDL-2's mean objective is 2.2399 billion units. Bioreactor shortage, patient loss, and expiry account for 44.10%, 43.53%, and 8.71%, respectively. The graph controller reduces the bioreactor-shortage component by 25.42 million units, or 2.57% of that component, which becomes a 1.13% improvement when divided by the complete objective. The component percentage is a mechanism measure; the complete-objective percentage remains the headline result.

This positive network checkpoint is driven primarily by advantage-filtered

Table 11: Cost decomposition for MDL-2 and Network AFR-GCN–DDPG in the abrupt-regional-shift holdout. The final column is the within-component percentage change. Components are grouped without changing the total-objective denominator used in Table 10.

Component	MDL-2 mean (10^6 units)	Share of total	AFR change (10^6 units)	Component change
Bioreactor shortage	987.891	44.10%	−25.420	−2.57%
Patient loss	975.120	43.53%	−0.352	−0.04%
Expiry	195.008	8.71%	−0.084	−0.04%
Urgency	6.118	0.27%	+0.641	+10.48%
Other operating components	75.742	3.38%	−0.178	−0.23%
Total weighted objective	2239.879	100.00%	−25.393	−1.13%

teacher distillation. Earlier unconstrained offline DDPG and TD3 updates weakened the policy or violated patient safeguards. The experiment therefore establishes value from the graph-aware AFR controller, but it does not by itself establish that online DDPG updates generated that value.

5.6.2. Conservative online-TD3 attribution screen

To test whether actual actor–critic updates add value beyond the frozen pretrained controller, we initialized matched graph and flat residual TD3 policies from their corresponding pretrained checkpoints and ran 100 online episodes for each of three seeds. Every run completed nonzero critic and actor updates with finite diagnostics. Final and frozen checkpoints were evaluated on identical 100-replication CRN holdouts.

Table 12: Final conservative TD3 outcomes under geographic regional demand drift. Values aggregate three training seeds and 100 paired holdout replications per seed.

Method	Objective (10^9 units)	Completion	Patients lost	Mfg. ineligibility
MDL-2	2.458807	0.490786	2211.86	0.073077
Final AFR-Flat-TD3	2.460346	0.490433	2213.75	0.073209
Final AFR-GCN–TD3	**2.456263**	**0.491636**	**2206.70**	**0.072685**

Table 13: Paired final-checkpoint TD3 comparisons. Objective differences are in millions of modeled weighted-objective units.

Comparison	ΔJ_{tot}	Relative gap	Paired 95% CI	Δ completion	Δ lost	Δ mfg. inelig.
Final AFR-GCN–TD3 vs. MDL-2	−2.544	−0.103459%	[−3.872, −1.131]	+0.000850	−5.16	−0.000392
Final AFR-Flat-TD3 vs. MDL-2	+1.539	+0.062595%	[+0.807, +2.387]	−0.000353	+1.89	+0.000132
Final AFR-GCN–TD3 vs. AFR-Flat-TD3	−4.083	−0.165951%	[−5.633, −2.436]	+0.001203	−7.05	−0.000524

The final graph policy improves on MDL-2 for each seed (−0.051936%, −0.139249%, and −0.119193%), deploys nonzero residuals in 73.50%, 47.63%, and 67.25% of evaluated decisions, and passes all pre-specified clinical noninferiority checks. The matched flat policies are more expensive than MDL-2 and fail clinical noninferiority for all three seeds. The final graph-versus-flat comparison is therefore statistically and clinically favorable.

The online-learning attribution is more limited. The pooled final-minus-frozen graph difference is -0.350 million units, with 95% interval $[-1.032, 0.155]$ million, and the GCN-minus-flat difference-in-differences is -0.485 million units, with interval $[-1.489, 0.173]$ million. Both point estimates favor online TD3 updates, but both intervals include zero. Thus the final graph controller outperforms MDL-2 and matched flat TD3 in this regional screen, while the independent incremental benefit of online TD3 over frozen pretraining is not yet established. A five-scenario zero-shot evaluation and additional training seeds are reserved for confirmation rather than being reported as completed evidence here.

6. Discussion and Conclusions

This study develops a patient-condition and geography-aware simulation of a 20-clinic PRM network, including identity-preserving pre-manufacturing specimen routing, and a residual GCN–DDPG controller that learns around a strong MDL-2 anchor. Advantage-filtered distillation identifies sparse state-dependent corrections, while checkpoint selection, residual-scale calibration, and fallback prevent an unsupported correction from being deployed. This structure preserves the interpretability of the anchor while allowing the graph actor to respond to patient pressure, transfer pipelines, regional disruptions, and demand-prior drift.

The numerical findings currently summarized below come from the earlier no-routing configurations and should be interpreted as controller evidence, not as evidence for or against specimen routing. The calibrated patient process places both ISO and MDL-2 inside a clinically interpretable manufacturing-period ineligibility band, and the weighted objective no longer rewards early patient loss. Under demand-prior drift, the formal five-seed result shows that AFR-GCN–DDPG improves on MDL-2 with a two-level confidence interval fully below zero while weakly improving all reported patient-facing metrics. The effect is small, and a matched flat residual policy also improves on MDL-2; consequently, this experiment supports anchored residual learning but not an independent graph advantage. The matched backbone comparison favors DDPG over TD3 for this replenishment-residual problem, which supports DDPG as the canonical proposed instantiation without implying that it is universally superior.

The graph-specific result emerges after the residual action is aligned with the network representation. Under an abrupt regional demand shift, Network AFR-GCN–DDPG significantly improves on both MDL-2 and a parameter-matched flat policy while improving completion service. The cost decomposition shows that the controller primarily reduces bioreactor-shortage penalties; reporting this component change alongside the full-objective percentage explains the mechanism without changing the headline denominator. A separate conservative online-TD3 screen likewise yields a final graph policy that outperforms MDL-2 and matched flat TD3 while satisfying clinical noninferiority. However, the pooled final-versus-frozen and graph-minus-flat difference-in-differences intervals both cross zero. We therefore distinguish the supported claim that the deployed graph-aware AFR controller creates value from the unresolved claim that online actor–critic updates independently create that value.

Several limitations remain. The formal demand-prior-drift DDPG experiment has five training seeds, but the network-action DDPG and conservative TD3 studies currently have three. The abrupt-shift graph checkpoint is dominated by advantage-filtered teacher distillation, and cross-scenario zero-shot robustness is not yet established. The full disruption-rate, robust/rolling MDL-2, pure-policy, SAC, and PPO comparisons still require a common locked budget. The clinical calibration target is literature aligned but is not fitted to patient-level data from a single indication, so sensitivity bands remain necessary. The patient-loss, expiry, urgency, and shortage coefficients are modeled decision weights rather than a complete accounting valuation; cost-weight sensitivity and external economic calibration are required before interpreting objective differences as realized monetary savings. External clinical data will also be needed to validate patient-risk, demand, disruption, and transfer-time assumptions before operational use. The routing-capable environment has passed implementation-level identity, conservation, lifecycle, and CRN tests, but the locked routing-primary teacher, smoke, pilot, and holdout stages have not yet produced results. Accordingly, this draft makes no empirical claim about the value of routing and does not reinterpret any legacy no-routing checkpoint as routing-enabled evidence.

Future Work This study motivates several extensions. Patient condition can be represented with longitudinal clinical covariates rather than the current eligibility and survival-risk summary. Raw-material quality, product quality, process failures, machine breakdowns, and operator availability can be added to the simulator. The immediate empirical priority is the locked routing-primary campaign: generate a fresh routing teacher, pass the two routing smoke runs, complete the six three-seed routing pilot runs, and evaluate GCN, matched flat, MDL-2, and frozen-pretraining controls under the same CRN protocol. No-routing training is reserved for a separately justified supplementary ablation rather than the default experiment. If online-learning attribution remains inconclusive, future training should use multi-scenario replay and a conservative, explicitly anchor-relative actor objective rather than merely increasing the episode budget. Longer-term experiments should test unseen clinic networks and transfer disruptions and should integrate production scheduling, product preparation and distribution, and treatment operations.

Data Availability

The data used in this study are generated from a simulation model. The simulation parameters and experimental settings are described in the manuscript. Code and additional materials are available from the corresponding author upon reasonable request.

Declaration of Competing Interest

The author declares that there is no known competing financial interest or personal relationship that could have appeared to influence the work reported in this paper.

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